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ORIGINAL ARTICLE

A high performance heterogeneous hardware architecture for brain computer interface

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Abstract

Brain–computer interface (BCI) has been widely used in human–computer interaction. The introduction of artificial intelligence has further improved the performance of BCI system. In recent years, the development of BCI has gradually shifted from personal computers to embedded devices, which boasts lower power consumption and smaller size, but at the cost of limited device resources and computing speed, thus can hardly improve the support of complex algorithms. This paper proposes a heterogeneous BCI architecture based on ARM + FPGA, enabling real-time processing of electroencephalogram (EEG) signals. Adopting data quantization, layer fusion and data augmentation to optimize the compact neural network model EEGNet, and design dedicated hardware engines to accelerate the network. Experimental results show that the system achieves 93.3% classification accuracy for steady-state visual evoked potential signals, with a time delay of 0.2 ms per trail, and a power consumption of approximately (1.91 W). That is 31.5 times faster acceleration is realized at the cost of only 0.7% lower accuracy compared with the conventional processor. The results show that the BCI architecture proposed in this study has strong practicability and high research significance.

Keywords Brain-computer interface (BCI) · Electroencephalogram (EEG) · Field-programmable gate-array (FPGA) · Hardware accelerator

1 Introduction

Brain–computer interface (BCI) is an emerging human–computer interaction (HCI) technology, which decodes the collected electroencephalogram (EEG) signals and converts them into instructions [1, 2]. These instructions are relayed to an external device to perform the specified operation [3]. BCI establishes a direct communication path between the human brain and external devices, thus eliminating the need for typical information transmission methods and improving the communication efficiency between human and machine, allowing individuals to directly control electronic devices, such as smart home appliances or auxiliary robots through their thoughts [4, 5]. In the past few decades, research on BCI has mainly focused on clinical applications, particularly enabling severely disabled individuals to interact with the environment. With the development of biomedical engineering and neuroscience technology, BCI has been used in more fields, such as the control of drones [6], aerospace applications [7], entertainment and games [8].

Today, biomedical science based on AI has a broad application scope, such as intelligent electrocardiogram (ECG) monitoring [9], electromyography (EMG) monitoring, disease prediction [10], and epilepsy detection [11]. AI-based classification algorithms are used for automatic diagnosis and classification. Compared with traditional signal processing algorithms, AI-based algorithms can help doctors improve the intelligence and accuracy of diagnosis or classification while automatically extracting and analyzing signal features [12]. At present, most AI applications of BCI systems are based on personal computers (PC), using offline remote computing for classification. With the

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increasing demand for consumer BCI products, there is a need for systems with higher integration levels, advances in embedded platforms and the availability of open source platforms have led to great interest in implementing BCI on mobile platforms [13]. Compared to PC devices, embedded devices excel in power consumption, comfort, and portability, and have more flexible application scenarios [14]. Meanwhile, using independent EEG processing devices can better protect users' personal privacy [15]. The application of AI algorithm effectively improves the accuracy of EEG signal processing, but if there is no reasonable planning and efficient hardware support, it will cause problems such as long system signal processing delay and high power consumption. The existing embedded system frameworks have limited the development of BCI system, the reasonable deployment of AI algorithms on embedded platforms has become a key issue.

At present, the mainstream embedded BCI (EBCI) structure is software structure, and over 90% of existing EBCI systems are implemented using software code from microcontroller units (MCU), such as ARM, Nios, microcomputers, etc. [16]. With the expansion of open source libraries, it's feasible to quickly complete the construction of BCI system prototypes. However, EEG signal processing requires advanced algorithms to filter, extract, and classify, which are usually based on complex mathematical operations. Software systems using MCUs can only access limited resources and adopt simple algorithms, which brings about problems such as decreased accuracy and long processing time. For EBCI systems based on hardware architecture, there are usually two hardware platforms, field-programmable gate array (FPGA) and application specific integrated circuit (ASIC). Priyadarshini proposed an effective epilepsy detection framework based on Optimized Adaptive Neuro Fuzzy Inference System (OANFIS) [17] and implemented the model on FPGA, which not only improves classification performance, but also reduces computational complexity in terms of area and power consumption. In ASIC design, dedicated biomedical processors have been proposed to accelerate AI-based biomedical processing algorithms [18]. Yu Xie [19] proposed an effective hardware acquisition and processing method for EEG biological signals, applied deep learning algorithm to the BCI system, and used the hardware system to accelerate the algorithm to obtain high accuracy and low delay. The pure hardware system can optimize the power consumption of the system based on the system characteristics, and its high computing power is highly suitable for processing high complexity algorithms [20]. However, for a complete EEG processing chain, it's very difficult to develop the hardware system and the development life cycle is too long, so cost and time for design significantly increased compared with those of the software architecture [21].

In this paper, an architecture of heterogeneous BCI system based on ARM + FPGA is proposed, which is developed by means of hardware and software co-design, and a compact convolutional neural networks (CNN) model EEGNet on the system for EEG feature extraction and classification is deployed. The architecture combines ARM processor and reconfigurable hardware FPGA. The ARM core is used to control the off-chip devices of BCI, and FPGA is used to compute complex data to achieve low power consumption and high resource utilization. For CNN model that need to be deployed, data augmentation is used in our design to improve the accuracy of the neural network, layer fusion is conducted to reduce the computational complexity of the network, and data quantization is adopted to compress the parameters of the network. These technologies can effectively reduce the computational complexity of neural network, and dedicated hardware acceleration engines are designed to improve the resource utilization of the system, which is conducive to the implementation on hardware. As a result, the accelerator uses fewer logic/arithmetic resources to complete CNN model deployments and has lower power consumption compared to other devices. The main contributions and innovations of our work are summarized below:

- Proposing a highly integrated BCI system architecture. Under this framework, EEG signal acquisition, preprocessing and CNN model reasoning are completed, which further optimizes resource utilization and power consumption, while showing higher system integration. These features expand the application scenarios of this design.
- Optimizing the compact CNN model EEGNet for hardware implementation. The optimization mainly includes three aspects: network quantification, layer fusion and data augmentation. Through these optimization methods, the size of the network can be effectively reduced without compromising the inference performance of the network.
- Setting up a high performance CNN hardware accelerator, and designing a dedicated pointwise convolution processing engine and a depthwise convolution processing engine for parallel processing operations, achieving lower resource usage and higher inference speed.
- Having implemented the designed system prototype on Xilinx ZYNQ xc7z020 FPGA, and the SSVEP signals are classified to evaluate the performance of the proposed system. Proving the reliability and practicability of this system framework through the comparison with desktop class devices and related work.

2 System overview

The system workflow built according to the proposed architecture is shown in Fig. 1a. The system is divided into online mode and offline mode. In the online mode, stimulation instructions and acquisition instructions are issued at the same time, and the corresponding module will call relevant peripherals and execute the program. The real-time data obtained from the acquisition unit will be sent to the preprocessing module for filtering and fast Fourier transform (FFT) processing at the same time, then it will be saved to the storage unit for offline analysis, After being preprocessing, the data is transmitted to the CNN classification model to output the classification results. For offline mode, the operation of the stimulation interface and data collection process are skipped, and the preprocessing will directly read the data from the storage unit for processing.

The proposed BCI system includes three devices: a LED stimulation panel (stimulus presentation), an EEG signal acquisition board (signal acquisition) and the main control module. The physical diagram of the system is shown in Fig. 1b.

3 Materials and methods

3.1 System architecture

Figure 2 shows the overall architecture of the system, which is mainly divided into on-chip and off-chip parts. The off-chip system includes a stimulation panel and a EEG signal acquisition board. The on-chip part is ARM + FPGA heterogeneous architecture proposed in this paper. It contains two parts: programmable logic (PL) and processing system (PS). We build the system design around this heterogeneous architecture based on the software and hardware collaborative workflow described in the previous section. In our design, the time-consuming complex algorithms in the system are processed by the hardware accelerator, and the rest is developed as embedded software on an embedded soft

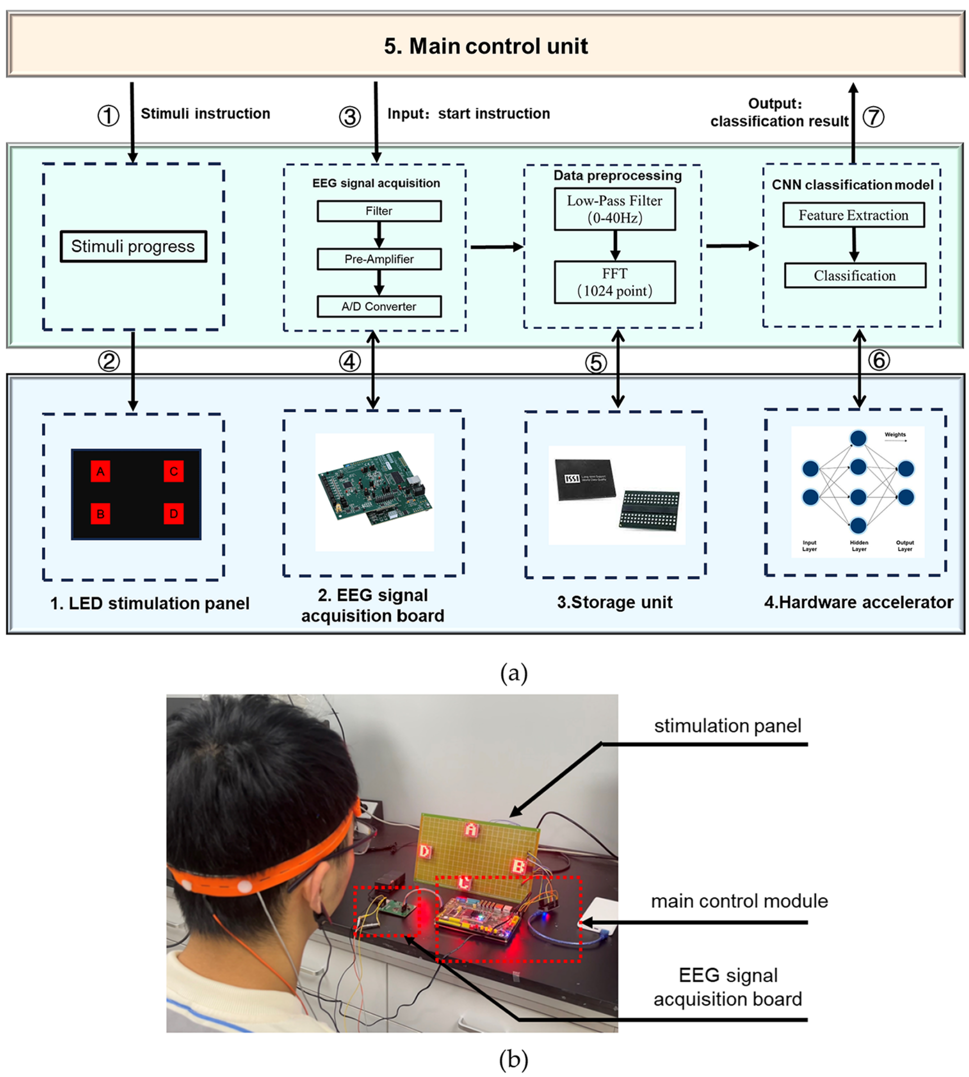


Fig. 1 **a** The main design of proposed BCI system. **b** The physical photo of the SSVEP-BCI system

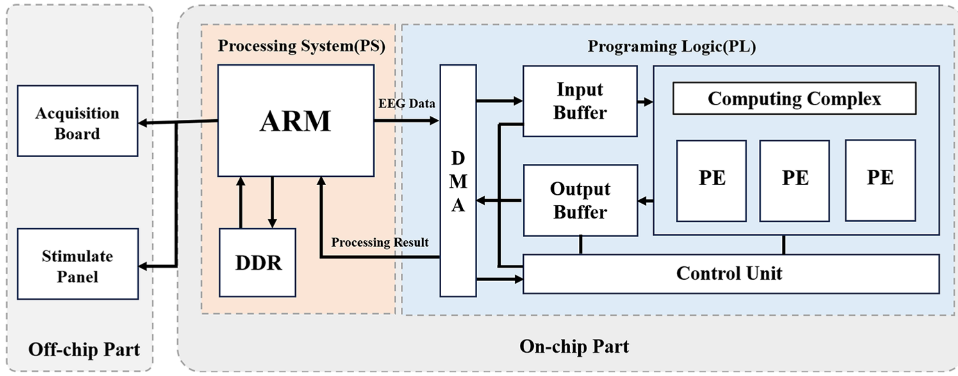


Fig. 2 Overall system architecture

core processor. This approach gives us more flexibility to optimize data processing time and power consumption. It is conducive to the rapid deployment and efficiency improvement of the whole design process of BCI system.

As the main control unit of the system, PS is composed of ARM core and external storage unit, which is responsible for controlling the off-chip system, on-chip data flow transmission control and accelerator configuration instructions. All CNN model data and collected signals are stored in external storage.

PL is the FPGA chip on which we complete the preprocessing of EEG data and the implementation of EEGNet model. We deploy complex computing modules, on-chip buffers, controllers, and Direct Memory Access (DMA) on it. The complex computing module consists of N parallel Processing Engines (PEs), which share the same input feature mapping and perform parallel computing for different output channels.

3.2 Signal preprocessing

Before the EEG signal is transmitted to the hardware accelerator, the input signal needs to be preprocessed in two steps in the preprocessing module. The module is mainly controlled by FPGA for data processing. The firmware consists of signal storage unit and data processing unit. The storage unit caches the signals collected each time, and transmits them to the pre-processing unit after units are full. The preprocessing unit is a low-pass filter with a cut-off frequency of 40 Hz, and then does a 1024-point FFT transformation to intercept the first 300 points as the input signal of the classification network. The data flow between the units is controlled by the transmission unit, and the original signal collected is transmitted to PC for comparing the errors generated by the system processing.

3.3 Network construction

EEGNet is a compact CNN model proposed by Vernon et al. [22], which is composed of three layers. The first layer is the Conv2D layer, which is used to filter EEG signals. The second layer is to learn the depth conversion of two dimensional blocks of spatial filters. The third layer is a Seprable-Conv2D block, which is used to extract the features of EEG decoding.

EEGNet uses DepthwiseConv2d and SeprableConv2d to ensure performance while effectively reducing the number of model parameters. This advantage is useful for embedded applications and specialized software. So we choose to deploy EEGNet in our BCI system. If the network is directly deployed on the hardware device, it will consume a lot of hardware resources and increase the signal processing time, so the direction of our model optimization is mainly to reduce the calculation work of the model. The model structure is shown in Fig. 3. The modifications we make to the original model include:

- Replacing the activation function of computationally complex ELU with a simple ReLU.
- Converting model parameters from floating point to fixed point.
- Fusing convolution layer and batch normalization layer.

For embedded platform algorithm mapping, parameter quantization is an effective method to reduce model computation. Traditional CNN usually uses 32-bit floating point numbers as calculation parameters and network weights for forward and back propagation, but the improvement of processing speed brought by using lower precision fixed-point numbers is far greater than the loss of network accuracy, and the storage space occupied by parameter numbers can be effectively reduced. We use different quantization bit widths to test the accuracy of the network. As shown in Fig. 4, when the quantization bit width reaches more than 16bit, the recognition accuracy can be almost the same as that of floating point numbers. The increase of quantization bit width will increase the calculation task of the system, so we believe that 16bit is the appropriate bit width.

The processing of batch normalization (BN) operations is included after each convolution layer of the model, which

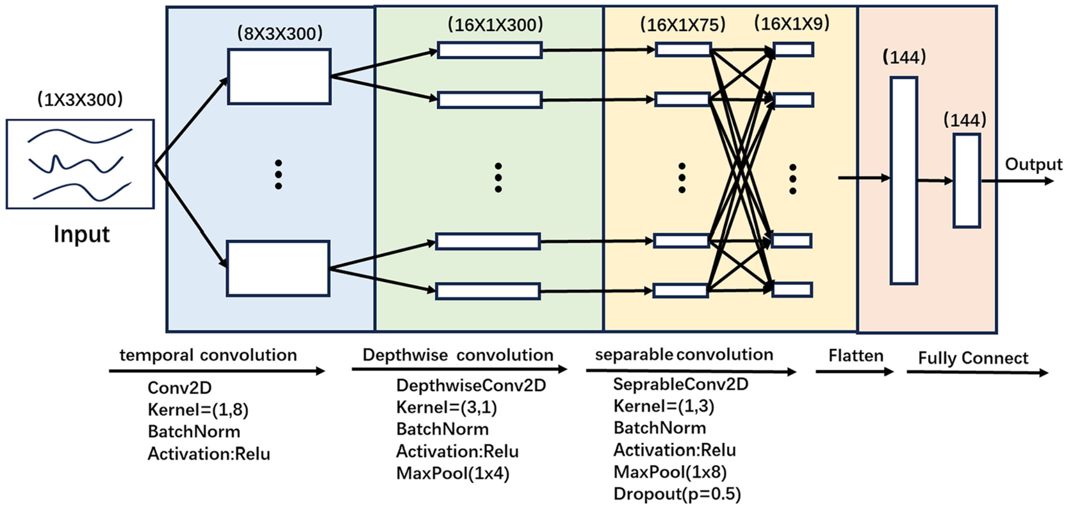


Fig. 3 Overview of EEGNet

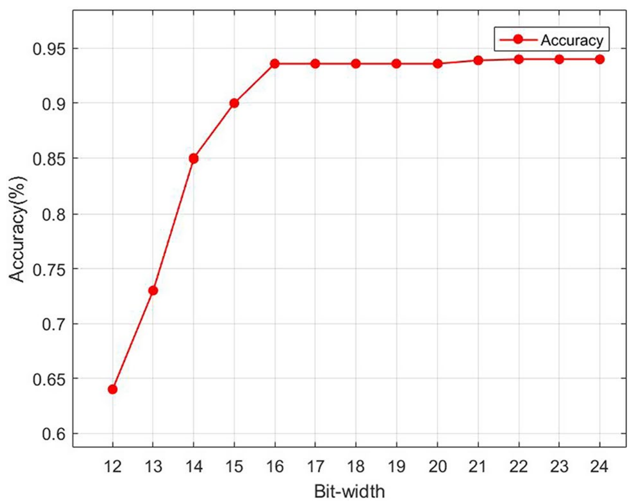


Fig. 4 The relationship between the classification accuracy and the bit width of the algorithm

is complicated to implement on FPGA hardware because it requires calculation of division and square roots. These operations are resource intensive and add a lot of latency. Therefore, we combine the BN layer with the convolution layer to reduce the number of model parameters and the cost of hardware resources, so as to improve the computing speed of the network model. The output of the convolution layer is shown in Eq. (1). Where **ω** is the weight of the convolution layer and b is the bias. The mathematical expression of the fusion processing of BN layer and convolution layer is shown in Eq. (2).

$$y_{conv} = \omega x + b \tag{1}$$

$$y_{bn} = \frac{\gamma}{\sqrt{Var[y_{conv}] + \varepsilon}} \cdot y_{conv} + \left(\beta - \frac{E[y_{conv}]}{\sqrt{Var[y_{conv}] + \varepsilon}} \cdot \gamma\right) \tag{2}$$

γ is the scale factor and β is the offset factor. Since the offset factor exists in the processing of BN layer, its own bias b in the convolution layer can be omitted. Equations (3) and (4) show the new weight and bias after the fusion of BN layer and convolution layer. Through the above derivation, it can be found that the fusion of BN layer and convolution layer effectively reduces the calculation amount of the network model, decreases the demand for hardware resources, and improves the operational efficiency of the entire model with almost no accuracy loss.

$$\hat{\omega} = \omega \cdot \left(\frac{\gamma}{\sqrt{Var[y_{conv}] + \varepsilon}}\right) \tag{3}$$

$$\hat{b} = \beta - \frac{E[y_{conv}] \cdot \gamma}{\sqrt{Var[y_{conv}] + \varepsilon}} \tag{4}$$

3.4 Data augmentation

Compared with traditional machine learning algorithms, deep learning methods rely heavily on high-quality data, otherwise insufficient amount of conventional data sets is prone to overfitting problems. To obtain enough high-quality data sets to train high-quality convolution kernel parameters is a key problem to be solved in CNN setup. Data augmentation can effectively improve this situation. Cheng et al. [23] conducted 10 kinds of data augmentation methods on EEG signals to train the model based on self-supervision, and achieved remarkable results. After testing, we choose to add noise and signal mixing to enhance the data.

We take the Tsinghua University (THU) 40 target SSVEP data set as the source data set for initial parameter training of the model [24]. The dataset includes 64-channel EEG data from 35 healthy subjects with 40 visual stimulus targets in the frequency range of 8–15.8 Hz, and an interval of 0.2 Hz. The phase difference between the two adjacent frequencies is 0.5π. The data matrix consists of 240 trials (40 targets × 6 blocks), each consisting of 1500 points of data across 64 channels. In each trial, the sampling frequency was 250 bps, and 6 s of data length (i.e. $6 \times 250 = 1500$ time points) were extracted, including 0.5 s before the stimulus began, 5 s before the stimulus began, and 0.5 s after the stimulus onset. The basic training set uses the three channels of O1, O2 and OZ with better classification performance and we used the time slice of the signal [2 s, 4 s] and the matrix size of each sample for the experiment is (1, 3, 300). The stimulus frequency is selected at 8.6 Hz, 10 Hz, 12 Hz and 15 Hz and divide the training set and the test set at the proportion of about 8:2, with a total of 640 samples and 160 samples for each stimulus. The enhanced training set using the additive noise method has 640 samples, adding independent and uniformly distributed Gaussian noise to the signal. There are 3840 enhanced training sets using the signal mixing method, which are generated by the basic training set signals superposing the signals of PO7, PO3, POz, PO4, PO6 and PO8 channels in the visual cortex regions respectively.

3.5 Hardware implement

3.5.1 Data flow overview

The hardware implementation is completed on FPGA, which aims to accelerate the CNN computing process as much as possible under the premise of making full use of on-chip resources. This goal is mainly achieved through configurable hardware structure and ram control. The acceleration flow is shown in Fig. 5. The whole accelerator system is carried out in three stages:

1. *Data loading* The initial network model parameters and EEG data are parsed through the AXI protocol path and the read/write control unit to obtain the start address and size of the required storage space, and the data stream is written into the On-chip cache.
2. *Data processing* This stage mainly consists of two parts. The first part is the convolution computing array, responsible for different modes of convolution computing. The convolutional computing array is composed of a large number of PEs. The detailed structure of PE is introduced in detail in Sect. 3.5.3. The second part is the post-processing module, the addition tree module is responsible for the accumulation and merging of the data after the convolution operation, the pooling module is used to complete the pooling calculation, and the non-linear module is used to perform ReLu calculation.

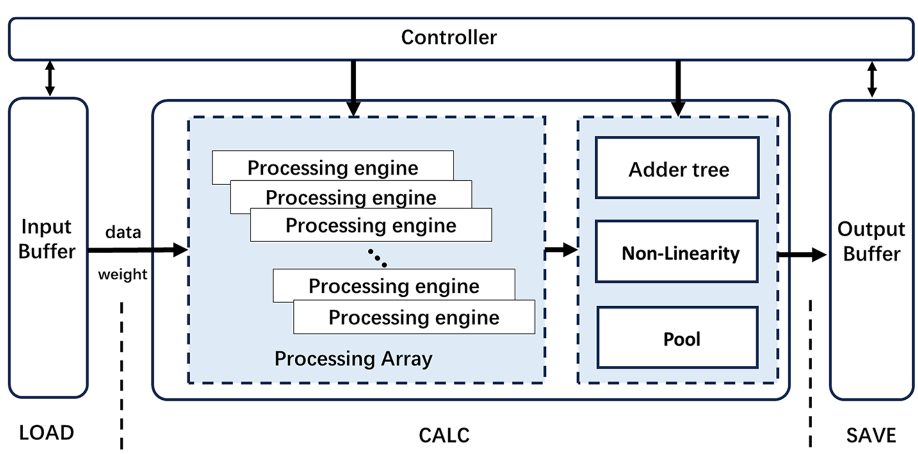
   When the data processing instructions arrive, the FPGA side calculation engine reads the data from the on-chip cache for inference tasks and the data processing is divided into several stages. Each module is responsible for the fixed calculation of the neural network layer. The detailed calculation sequence is described in Sect. 3.5.2.

3. *Data saving* After the accelerator completes the calculation task, the read/write control unit recalls the output feature map to the PS terminal.

3.5.2 Computation sequence optimization

We improve the parallelism of data processing to optimize performance by means of intra-layer parallelism and inter-layer pipelining. Intra-layer parallelism means that the computation of all channels in a convolution layer is carried out in parallel, and inter-layer pipelining means that the next layer can read part of the results of the previous layer for calculation without waiting for the calculation of the whole

Fig. 5 The data processing flow: load, calculate, save



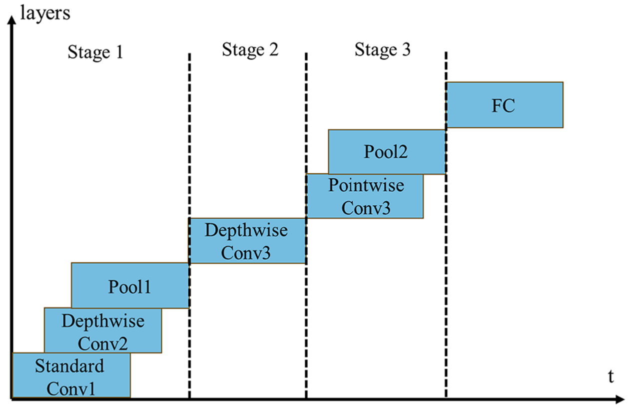


Fig. 6 Data acceleration process

layer. The optimized calculation timing diagram is shown in Fig. 6. The whole accelerated reasoning process is divided into three stages. The first stage includes the first layer of standard convolution, the second layer of depthwise convolution, and the first layer of pooling; the second stage only calculates the third layer of depthwise convolution with high parallelism requirement; the third stage includes the third layer of pointwise convolution kernel; the second layer of pooling; and the calculation of the fully connected layer is performed after all the previous calculations are completed. In the absence of interlayer parallelism, all feature maps of each layer need to be first written to memory and then read by the next layer for calculation, resulting in a large inference delay.

3.5.3 Processing engine

Processing engine is the basic unit for convolutional calculation. In EEGNet, three kinds of convolution are mainly used: standard convolution, depthwise convolution (DWC) and pointwise convolution (PWC). Based on the characteristics of FPGA parallel computing, we design two types of PEs for the used CNN model: The PWC accelerator and the optional DWC accelerator which increases the reusability of the hardware architecture so that different neural networks can also use this architecture.

For DWC accelerator, its structure is shown in the Fig. 7a, here are two register groups in the process of the DWC calculation, one for storing the input data and the other one for storing the weight parameters. The data in the register is updated through the shift operation. We design variable register size to adapt to different convolution core sizes. At the same time, because the input channel of the standard convolution layer is a single channel, we can use DWC accelerator in the calculation of standard convolution. For PWC accelerator, its structure is shown in the Fig. 7b, since the number of convolution kernel channels is the same as the number of input data

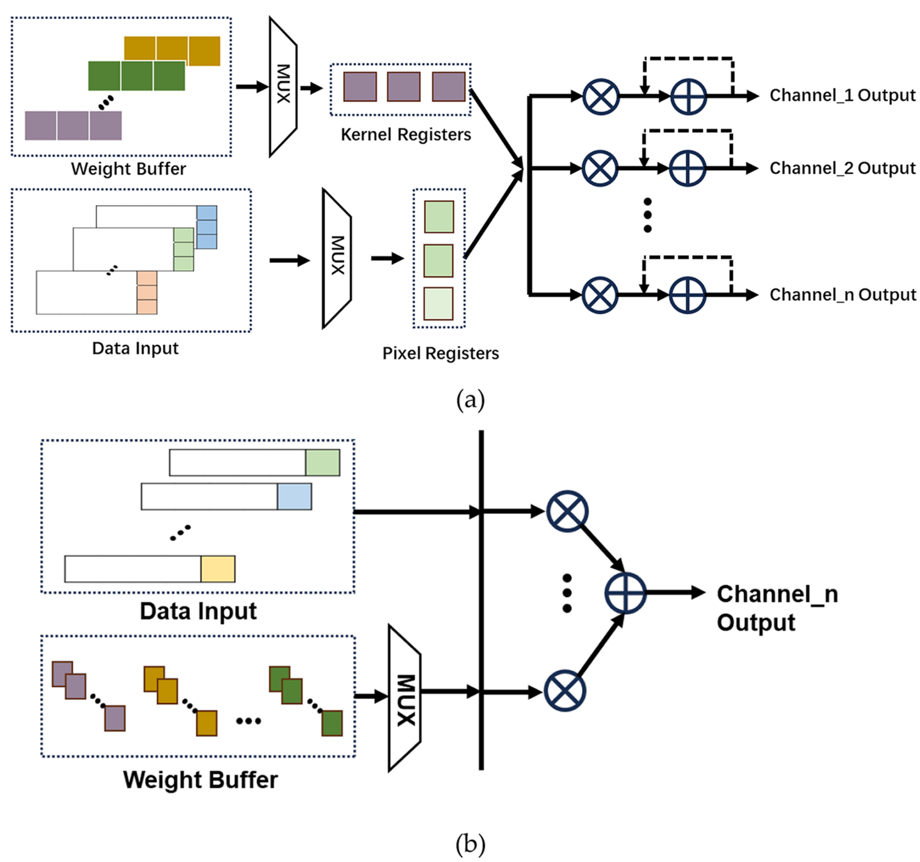


Fig. 7 **a** Depthwise convolution accelerator architecture. **b** Pointwise convolution accelerator architecture

channels, the data on each input channel will be convolved with multiple convolutional cores. Therefore, the PWC operation mainly consists of three register groups, a register for storing input data, a register for storing weight parameters and a temporary register. During the PWC operation, the output result of each channel operation is stored in the temporary register, and the result is accumulated through the addition tree to output the convolution sum.

3.6 Experimental design

We design offline experiments to verify the optimization effect of the CNN model, and design online experiments to verify the performance of our BCI system. In order to facilitate calculation, the output of offline EEG data and online classification results are transmitted to the computer synchronously through communication unit.

In the offline experiment, participants included 10 healthy subjects (8 males and 2 females, ages 21–27). All participants had normal or corrected vision. Informed consent forms were read and signed by all participants. According to the international 10–20 system, three electrodes were placed on O1, Oz and O2, with double mastoids as reference and ground electrodes. Subjects were seated in a comfortable chair 60 cm away from the stimulation panel, and in order to avoid any random errors and maintain the consistency of the system, a total of 20 rounds of tests were conducted. In each round of the experiment, participants were asked to look at a single target during the flicker of the target color block until the flicker stopped, and to finish the round after completing the staring of all four targets. Each round of targets had a flicker duration of 10 s, and in addition, they were given five second intervals to shift their gaze between two consecutive targets. During the experiment, EEG signals were recorded without any pretreatment. All four targets were flashing until the end of each trial. To prevent visual fatigue, subjects were given a 2-min rest after each round of the test.

In the online experiment, participants were asked to focus on the corresponding color block and blink as little as possible. We asked participants to gaze at random targets through voice prompts, with the duration of each stimulus target set to 3 s. A total of 16 rounds of tests were performed with an interval of 0.5 s between each stimulus, and make sure all stimulus targets are traversed at least once. Finally, the system performance was evaluated by ITR, which is an important index to evaluate the classifier performance of BCI system. The formula is:

$$ITR = \left\{ \log_2 N + P \log_2 P + (1-P) \log_2 \left[\frac{(1-P)}{(N-1)} \right] \right\} \cdot \left\{ \frac{60}{T} \right\} \tag{5}$$

Table 1 Resource utilization of the accelerator

FPGA resource	Used	Available	Used rate %
LUT	26,768	53,200	50.31
Flip flop	26,768	106,400	27.43
DSP	181	220	82.27
BRAM	78.5	140	56.07

Table 2 Effect of model optimization strategy on accuracy (%)

	EEGNET		DE-EEGNet		DE-EEGNet Plus
Activation	ELU	ReLU	ReLU	ReLU	ReLU
Layer fusion	X	√	√	√	√
Quantization	X	X	X	16bit	24bit
LUTs	–	–	–	26,768	48,237
Accuracy	89.2	89.1	93.7	93.5	93.7

where N is the total number of stimuli, P is the accuracy rate, and T is the time required to recognize a command.

4 Results

4.1 Resource consumption

We used Xilinx Zynq xc7z020 to implement the designed system architecture, which uses ARM + FPGA SoC technology to integrate dual-core ARM Cortex-A9 and FPGA on one chip. Table 1 shows the main hardware resource utilization of our design. The utilization of Look Up Tables (LUTs), Flip-Flops (Flip Flops), Bram and Digital Signal Processors (DSPs) in use were 26,768, 26,768, 181 and 78.5, respectively. The Brams were the on-chip memory resource of FPGA, and DSP blocks were mainly used for complex network calculation. For embedded platforms with limited resources, we found a balance between accelerated performance and resource usage, successfully deployed the network on the FPGA, and fully utilized the resources.

4.2 Network optimization performance

We compared the classification accuracy of the EEGNet, the EEGet after ReLU activation and layer fusion, the data augmentation EEGNet (DAEEGNet), and the DAEEGNet using different fixed-point quantization strategies. Table 2 reports the average accuracy and standard deviation during operation. Compared with the original EEGNet, the impact of network modification (i.e. using RelU instead of ELU and layer fusion) on classification accuracy is 0.1%, which can be ignored. In addition, the utilization of data augmentation yields a 4.6% increase from to the initial, which is a superior outcome. It proves the effectiveness of the data augmentation strategy we used. At the same time, we compared the two quantization bit widths of 16bit and full precision 24bit, we can observe that the accuracy of 16bit quantization is comparable to that of the 24bit quantization method, and the resource consumption decreases significantly, indicating that the selection of the 16bit quantization width is a reasonable option.

4.3 Accelerator performance

As shown in Table 3, we used test set data on different platforms to compare the performance of our system with that of desktop-level hardware to demonstrate the validity of our design. We perform CNN model inference calculations in the software using the PyTorch framework with CUDA 11.0 on Nvidia RTX 4060 GPU and INTEL I7-14700KF CPU. Thermal design power (TDP) values for CPU and GPU are 35 W and 115W respectively. According to the power report provided by the Zynq design Suite, the total on-chip power of our hardware architecture is only 1.91 W. Therefore, our design has advantages in power consumption and is suitable for use in embedded scenarios. In terms of reasoning speed, we designed the reasoning speed of 0.2 ms, which is 31.5 times of CPU and 7.5 times of GPU. For the CNN model deployed on FPGA, the accuracy rate can reach 93.3%. We can see that the accuracy of the model on the FPGA platform is slightly lower than that of the PC-side device, which is about the same as expected, and we think this slight decrease in accuracy is acceptable.

4.4 System performance

When using the 3-s time window for online experiments, according to the test results shown in Table 4, we found that the system proposed in this study exhibited stable accuracy and high ITR. In 10 subjects, the average accuracy reached 94.4%, while the average ITR reached 28.54 bit/min. These results show that our system can maintain high accuracy in processing tasks and has high information transmission rate. Compared with the results of off-line experiment, there's not much difference between the results of on-line experiment and off-line experiment. This means that our system can maintain the performance equivalent to that of offline experiments in actual use, which shows that the system is real-time and reliable, thus is suitable for applications in real scenes.

Table 3 Performance comparison between different platforms

Platform	CPU	GPU	FPGA
Device	INTEL I7-14700KF	Nvidia RTX 4060	ZYNQ xc7z020
Bit-width	Float32/Int16	Float32/Int16	Int16
Accuracy (%)	93.7/93.6	93.7/93.6	93.3
Frequency (MHz)	3400	1320	50
Latency (ms)	6.3/5.7	1.5/1.3	0.2
Power (W)	35	115	1.91

Table 4 Accuracy (%) and ITR (bits/minute) of SSVEP-based BCI Systems

Subject	ITR (bit/min)	Accuracy (%)
S1	34.28	100
S2	24.86	87.50
S3	34.28	100
S4	34.28	100
S5	24.86	93.75
S6	24.86	87.5
S7	34.28	100
S8	28.83	93.75
S9	18.59	81.25
S10	34.28	100
Average	28.54	94.4

In addition, Table 5 summarizes the system comparison between other SSVEP based BCI systems and our proposed system. In terms of accuracy, using EEGNet for classification has more advantages than using traditional algorithms. However, in terms of ITR, our system is slightly lower than some of the other designs. This is mainly because our system uses fewer stimuli. This can be one of the directions for our future improvement, increasing the diversity of stimuli to improve the ITR of the system. At the same time, our system uses fewer electrodes, which is significant to improve the portability and comfort of the system.

In summary, our system performs well in terms of accuracy and ITR, and has performance comparable to offline experiments. Compared to other SSVEP-based BCI systems, we are at a high level in terms of accuracy, but there is still room for improvement in terms of ITR. Increasing the number of stimuli and optimizing the portability of the system will be the way forward.

Table 5 Comparison with related work

Reference	Method	ITR (bit/min)	Accuracy (%)	Channel	Processer	Number of targets
Shyu et al. [25]	Phase Code	24	89	1	FPGA	4
Guneysu et al. [26]	FFT	–	75	14	PC	4
Damaševičius et al. [27]	LDA/SVM	–	79.3/80.5	16	PC	3
Wang et al. [28]	CCA	4.6	83.3	16	PC	4
Lin et al. [14]	FFT	47.93	91	1	FPGA	6
Ours	EEGNet	28.54	94.4	3	FPGA	4

5 Conclusions

In this paper, we propose a integrated high-performance BCI system architecture, which uses ARM-FPGA heterogeneous architecture to realize efficient processing of the complete EEG processing chain, improving system performance and speeds up the efficiency of interconnection between electronic devices, avoiding the drawbacks of traditional embedded systems that are difficult to deploy AI algorithms efficiently. The ARM part is responsible for the control of off-chip devices and the control of on-chip data flow, while the FPGA part is responsible for signal processing and hardware acceleration, maximizing the advantages of heterogeneous architecture. We have optimized a compact CNN model EEGNet to greatly reduce its computation complexity while avoiding a decrease in accuracy, which is beneficial for hardware deployment. We have designed dedicated PEs to further improve the system's resource utilization and acceleration performance. We can also determine the number of PEs to control the signal processing speed according to the complexity of the algorithm and different hardware resources.

We implemented the designed system architecture on Xilinx Zynq platform and classified SSVEP signals. The average accuracy of the system was 93.3%, the processing delay of each trail of EEG signal was 0.2 ms, and the overall power consumption was only 1.91W, which met the requirements of high precision, low power consumption and low delay of EBCI. We believe that the proposed system architecture can be also applied to the algorithm deployment of other paradigms such as MI, P300, etc. At the same time, due to the collaborative architecture design of hardware and software, the system has better versatility and flexibility and can be easily extended to other electronic equipment control such as robotic arms, typewriters, drones, which is beneficial to promote the application of BCI in various fields.

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Declarations

Conflict of interest All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethical approval All subjects were informed and agreed to participate in the experiment.

Consent to participate All authors declare their consent to participate in the paper. Informed consent was obtained from all individual participants included in the study.

Consent to publish All authors declare their consent to publication.

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